Gastroesophageal Reflux Disease in the Child and Adolescent: Making the Diagnosis. The Empiric Proton Pump Inhibitor Trial, pH-Impedance Monitoring by the Lyon Consensus, Endoscopy, and the Limits of Paediatric Norms.

A fourteen-year-old girl comes to clinic with burning behind her breastbone two or three times a week, worse after late meals. Her paediatrician started a proton pump inhibitor, she felt better within a fortnight, and she has now been on it for eighteen months because every time she stops, the burning returns. Her parents ask a simple question: does she have reflux disease, and will she need this medicine for the rest of her life? The honest answer is that nobody has yet shown that she has reflux disease at all. She has a symptom, and she had a response to a drug, and neither of those is proof.

The clinical question of this chapter is therefore this. When is heartburn in a child or adolescent really gastroesophageal reflux disease, abbreviated GERD, and how do we prove it? The main idea is that in older children and adolescents with typical symptoms, a short empiric trial of a proton pump inhibitor, abbreviated PPI, is reasonable, but a response to it does not prove GERD. Proof comes from one of two places: erosive oesophagitis of Los Angeles grade B or worse at endoscopy, or abnormal reflux monitoring off acid suppression. And the numerical thresholds we use for reflux monitoring in children are borrowed from adults or set by consensus in sick children, because there are no healthy paediatric controls. The clinical stake is large. Most adolescents with heartburn and a normal endoscopy turn out not to have excess acid exposure, and without testing they can stay on acid suppression for years.

A note on sources. The paediatric framework is the 2018 joint clinical practice guideline on paediatric gastroesophageal reflux by Rosen, Vandenplas and colleagues, written for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. The reflux monitoring thresholds come from the Lyon Consensus, first published in 2018 and updated as Lyon Consensus 2.0 by Gyawali and colleagues in the journal Gut in 2024. The adult guidelines are the 2022 American College of Gastroenterology guideline by Katz and colleagues, and the 2022 American Gastroenterological Association clinical practice update by Yadlapati and colleagues. Paediatric data come from the European paediatric impedance working group protocol of 2012, the German Paediatric Impedance Group study by Pilic and colleagues in 2011, a 2024 paediatric baseline impedance study by Pop and colleagues, paediatric endoscopy series by

Gilger, El-Serag and Nguyen, and a 2017 study of reflux phenotypes in children by Mahoney, Nurko and Rosen in Boston.

Part one. One symptom, four mechanisms.

Before choosing a test, it helps to know what we are trying to separate. Heartburn in an adolescent can arise from four different mechanisms. The first is erosive oesophagitis: acid injures the lining of the oesophagus enough to cause visible breaks in the mucosa. The second is non-erosive reflux disease, abbreviated NERD, in which the lining looks normal but monitoring shows that the oesophagus is exposed to acid for longer than normal. The third is reflux hypersensitivity, in which acid exposure is normal, but the symptoms still line up in time with ordinary, physiological reflux events, so the oesophagus is unusually sensitive. The fourth is functional heartburn, which the new paediatric Rome criteria call reflux-negative oesophageal pain, in which acid exposure is normal and the symptoms bear no relation to reflux events at all.

Only the first two are GERD in the strict sense. The last two are disorders of gut and brain interaction, in which acid is at most a trigger. They respond to PPIs inconsistently, and they often do better with neuromodulating medicines and behavioural therapy. The key point is that an empiric PPI trial cannot tell these four apart. Patients in all four groups can feel better on a PPI, whether through acid suppression, through placebo response or through natural fluctuation. Reflux monitoring and endoscopy can tell them apart, and that is their real job in the adolescent.

Part two. The tests that do not diagnose GERD.

The 2018 paediatric guideline is unusually clear about what not to do. The panel agreed unanimously that barium contrast studies, ultrasound, nuclear scintigraphy, oesophageal manometry, salivary pepsin and endoscopy on its own should not be used to diagnose GERD. The accuracy figures cited in the guideline show why. If you picture each test as a pair of bars, one for sensitivity and one for specificity, ultrasound has a long sensitivity bar of about ninety-five percent and a tiny specificity bar of about eleven percent, because it sees reflux, which almost every child has, rather than reflux disease. Scintigraphy has a sensitivity of about sixty-nine percent and a specificity of about seventy-eight percent. Salivary pepsin is about seventy-two percent and seventy-one percent. The reflux index on pH monitoring alone is about fifty percent sensitive and eighty-two percent specific. And the empiric PPI trial, using pooled adult data, is about seventy-eight percent sensitive and fifty-four percent specific. None of these is a good test.

Some of these investigations still have a role, but for different questions. Barium and ultrasound can exclude anatomical problems such as malrotation or pyloric stenosis. Manometry is useful when a motility disorder such as achalasia is suspected. And the guideline

notes that the appearance of the larynx, and the lipid-laden macrophage index from airway samples, do not correlate with pathological reflux, so they should not be used to diagnose it.

Part three. The empiric PPI trial: a trial, not a test.

Where does the empiric PPI trial fit? The 2018 paediatric guideline makes three recommendations. First, in children with typical symptoms, meaning heartburn, retrosternal pain or epigastric pain, a four to eight week trial of a PPI may be used as a diagnostic test; ninety percent of the panel agreed. Second, a PPI trial should not be used as a diagnostic test in infants, with eighty percent agreement. Third, a PPI trial should not be used as a diagnostic test for extra-oesophageal symptoms such as cough, wheeze, hoarseness or dental erosion, with unanimous agreement. The phrase typical symptoms matters, because children generally cannot describe heartburn reliably until about eight years of age.

The adult documents say much the same. The 2022 American College of Gastroenterology guideline recommends an eight-week trial of a once-daily PPI before a meal for classic heartburn and regurgitation without alarm features, and then an attempt to stop the drug if it works. The 2022 American Gastroenterological Association update suggests four to eight weeks of a single daily dose, and advises up-front testing rather than an empiric PPI for isolated extra-oesophageal symptoms.

In practice, for adolescents, the doses on the American drug labels are as follows. Omeprazole is labelled from one year of age by weight: five milligrams from five to under ten kilograms, ten milligrams from ten to under twenty kilograms, and twenty milligrams at twenty kilograms or more. Esomeprazole for twelve to seventeen year olds is twenty milligrams daily for symptomatic GERD, and twenty or forty milligrams for erosive oesophagitis. Lansoprazole for the same age group is fifteen milligrams for GERD and thirty milligrams for erosive disease. Pantoprazole from five years of age is twenty milligrams between fifteen and forty kilograms, and forty milligrams above forty kilograms. All of these should be taken about thirty minutes before breakfast, because PPIs only block acid pumps that are active, and labels should be checked before prescribing.

Part four. Why a response barely moves the needle.

The weakness of the empiric trial becomes obvious when you do the arithmetic. In 2004, Numans and colleagues pooled fifteen adult studies of a short PPI course as a test for GERD and found a sensitivity of seventy-eight percent and a specificity of fifty-four percent. That gives a positive likelihood ratio of about 1.6 to 1.9. A likelihood ratio tells you how much a test result changes the odds of disease, and anything below two is weak. Using a likelihood ratio of about 1.7 for a response and about 0.41 for no response, an adolescent who starts with a fifty percent chance of GERD has about a sixty-three percent chance after responding, and about a twenty-nine percent chance after not responding. If you picture these as three bars, the

bar rises only modestly after a response and falls only modestly after no response. A response moves the probability by about thirteen percentage points.

Why so weak? Placebo response, the natural ebb and flow of symptoms, and the fact that reflux hypersensitivity and even some functional pain can improve on a PPI. The paediatric guideline also notes that more than half of patients with typical symptoms may not respond to acid suppression at all. Lyon Consensus 2.0 puts it bluntly: a response to an empiric PPI trial alone is insufficient for a conclusive diagnosis of GERD. And accuracy has never been measured in adolescents; every one of these figures comes from adults.

Part five. Reflux monitoring off PPI: six, four and the grey zone.

The central number in reflux monitoring is the acid exposure time, abbreviated AET: the percentage of the twenty-four-hour recording during which the pH in the distal oesophagus is below four. Lyon Consensus 2.0 sets three bands for a patient with unproven GERD studied off acid suppression. An acid exposure time below four percent is physiological. An acid exposure time above six percent is conclusive for GERD, with ninety-four percent agreement among the panel. Between four and six percent is the grey zone, which needs further context and additional data.

Three other measurements help to break ties in that grey zone. The first is the number of reflux episodes in twenty-four hours, which impedance monitoring counts regardless of acidity. Fewer than forty episodes is evidence against GERD, more than eighty is supportive, and forty to eighty is inconclusive on its own. The second is the mean nocturnal baseline impedance, abbreviated MNBI, which measures the electrical resistance of the oesophageal lining during sleep. A healthy lining has high resistance, while a lining damaged by long-term acid becomes leaky and has low resistance. A value below 1,500 ohms supports GERD, and a value above 2,500 ohms argues against it. The third is symptom association. The symptom index, abbreviated SI, is the percentage of symptoms that follow a reflux event, and a value of fifty percent or more is positive. The symptom association probability, abbreviated SAP, is a statistical test of whether that association could be due to chance, and a value of ninety-five percent or more is positive. Picture these as four horizontal bars, each divided into a green band for normal, an orange band for inconclusive and a red band for abnormal. Only the acid exposure time bar is diagnostic. The others shift the interpretation in the grey zone, and symptom association raises confidence, but it is never diagnostic on its own.

Part six. What changed in Lyon 2.0: B is now enough.

Lyon 2.0 made several changes that matter in practice. Los Angeles grade B oesophagitis, which in 2018 was borderline, is now conclusive evidence of GERD. Grade A remains borderline, because it is seen in five to seven and a half percent of healthy adults. A hiatus hernia now counts only as supportive evidence. For a patient with proven GERD who still has

symptoms on an optimised PPI, Lyon 2.0 introduced on-therapy criteria: an acid exposure time above four percent together with more than eighty reflux episodes on pH-impedance defines actionable refractory GERD. For wireless capsule monitoring, ninety-six hours is preferred. An acid exposure time above six percent on two or more days is diagnostic; below four percent on all days with negative symptom association excludes GERD; and below four percent on all days with positive symptom association means reflux hypersensitivity. Some older measures were retired, including an index of swallow-induced waves after reflux, which became a research tool, and routine biopsy.

Lyon 2.0 also defined actionable GERD: conclusive endoscopy findings or abnormal reflux monitoring, together with compatible troublesome symptoms. Actionable means that the evidence is strong enough to justify long-term acid suppression, escalation of treatment, or an intervention such as fundoplication. There is, however, an important paediatric caveat. Lyon 2.0 was written by twenty-one adult experts and does not mention children at all. Its numbers are a reasonable frame for adolescents, but they are not a validated paediatric norm.

Part seven. Children have no normal controls.

The deeper problem in paediatric reflux monitoring is that we do not know what normal looks like. The 2018 guideline states directly that obtaining data in healthy controls is not ethically feasible: nobody puts a nasal catheter into a healthy child for twenty-four hours. Every paediatric value therefore comes from symptomatic children. For acid exposure, the 2018 guideline text defines an abnormal reflux index as more than ten percent of the time below pH four in infants under one year, and more than five percent above one year. The 2009 guideline had treated values up to seven percent as normal over one year of age. For reflux episodes, the German Paediatric Impedance Group, which studied seven hundred children from three weeks to sixteen years of age, defined an abnormal study as more than one hundred episodes a day under one year and more than seventy a day from one year, and the authors said themselves that these cut-offs were chosen arbitrarily. For symptom association, the 2012 European paediatric impedance working group protocol uses a symptom index of at least fifty percent and a symptom association probability above ninety-five percent, with a two-minute window agreed by consensus. For baseline impedance there is no validated paediatric cut-off, and typical values in children are between two thousand and four thousand ohms. Wireless capsule monitoring is rarely used in children, and most paediatric studies are twenty-four-hour catheter recordings.

Part eight. Impedance sees what pH misses.

Why combine impedance with pH at all? Because impedance detects the movement of fluid up the oesophagus whatever its acidity. That matters in children, who often reflux weakly acidic gastric contents after feeds, and in anyone on acid suppression. In the German study of seven hundred children, 270 studies were abnormal. Of these, forty-five percent, 120 studies, were

abnormal only on impedance, thirty-seven percent, 101 studies, were abnormal on both, and eighteen percent, forty-nine studies, were abnormal only on pH. In other words, nearly half of the abnormal studies would have been missed by pH monitoring alone. The rate of abnormal studies also varied by referral reason: about half of studies done for neurological indications were abnormal, forty percent of those for respiratory indications, and thirty-five percent of those for gastrointestinal indications.

The 2018 guideline supports pH-impedance, with eighty-nine percent agreement, to correlate persistent troublesome symptoms with acid and non-acid reflux events, and to clarify the role of reflux in oesophagitis. It supports pH-impedance to judge whether acid suppression is adequate with seventy-eight percent agreement. But its use to separate non-erosive disease, reflux hypersensitivity and functional heartburn reached only seventy-three percent agreement, just below the panel's seventy-five percent threshold for consensus.

Part nine. Baseline impedance: the mucosa keeps a record.

Mean nocturnal baseline impedance is attractive because it reflects the state of the lining over weeks rather than the events of a single day. In 2024, Pop and colleagues studied forty-nine children. A cut-off of 1,457 ohms separated non-erosive GERD from other groups, with an area under the receiver operating curve of 0.864, a sensitivity of eighty-nine percent and a specificity of seventy-five percent. A cut-off of 2,563 ohms identified functional heartburn, with an area under the curve of 0.802, a sensitivity of eighty-four percent and a specificity of eighty-one percent. These values land remarkably close to the adult Lyon thresholds of 1,500 and 2,500 ohms, which is reassuring. But this was one centre and forty-nine children, so in practice baseline impedance should be used to break ties in the grey zone, not as a stand-alone diagnosis.

Part ten. Endoscopy and the Los Angeles classification.

The Los Angeles classification, validated by Lundell and colleagues in 1999, grades mucosal breaks in the distal oesophagus. Grade A means one or more breaks no longer than five millimetres that do not extend between the tops of two mucosal folds. Grade B means at least one break longer than five millimetres that still does not run continuously between the tops of two folds. Grade C means breaks that are continuous between the tops of two or more folds but involve less than three quarters of the circumference. Grade D means breaks involving three quarters of the circumference or more. Under Lyon 2.0, grades B, C and D are conclusive for GERD, while grade A is borderline. Biopsy-proven Barrett's oesophagus and peptic stricture are also conclusive. The American College of Gastroenterology asks for Barrett's longer than three centimetres with intestinal metaplasia, and the American Gastroenterological Association uses three centimetres or more.

Timing matters. In a patient whose GERD is not yet proven, Lyon 2.0 recommends endoscopy at least two, and preferably four, weeks after stopping the PPI, because acid suppression heals grade A and B lesions and hides the evidence. Interestingly, the 2018 paediatric guideline does not cite the Los Angeles classification at all, even though paediatric endoscopists use it routinely.

Part eleven. Erosions are the exception in children.

Erosive disease is much less common in children than in adults. In a large American multicentre series published by Gilger and colleagues in 2008, erosive oesophagitis was found in 12.4 percent of roughly seven thousand two hundred paediatric endoscopies. In a 2002 study by El-Serag and colleagues of 402 neurologically normal children with GERD symptoms, erosive oesophagitis was found in 34.6 percent, with no Barrett's oesophagus. The 2018 guideline describes a range from fifteen to seventy-one percent across studies of children with troublesome symptoms, a spread that mostly reflects referral bias. Barrett's oesophagus is rare: in a 2011 series of 840 children by Nguyen and colleagues, Barrett's was suspected in 1.43 percent but confirmed histologically in only one child, 0.12 percent. If you picture these as bars, erosive disease is a modest bar, and Barrett's barely leaves the baseline.

The paediatric guideline recommends endoscopy with biopsies to assess complications of GERD, when mucosal disease is suspected, and before escalating therapy, with ninety percent agreement. In practice this means children with dysphagia, painful swallowing, bleeding, weight loss, failure of a PPI trial, or before long-term therapy or surgery.

Part twelve. Biopsies exclude more than they include.

What do biopsies add? The changes of so-called microscopic oesophagitis, such as thickening of the basal cell layer and lengthening of the papillae, are non-specific and do not diagnose GERD. A normal biopsy does not exclude GERD, because non-erosive disease is common. Lyon 2.0 removed routine biopsy from the diagnostic criteria. The main value of biopsy in a child is to find something else, above all eosinophilic oesophagitis, defined by fifteen or more eosinophils per high-power field in the right clinical setting. The 2018 guideline notes that up to eight percent of children investigated for extra-oesophageal symptoms have eosinophilic oesophagitis, and finding it changes treatment entirely. So biopsies should be taken from the proximal and distal oesophagus at every paediatric endoscopy for reflux symptoms, ideally off PPI.

Part thirteen. Phenotypes: most heartburn with a normal scope is not acid.

The most clinically important paediatric study for this chapter comes from Boston. In 2017, Mahoney, Nurko and Rosen studied forty-five children aged five years or older with heartburn

and a normal endoscopy, using pH-impedance. Twenty-seven percent had non-erosive GERD, twenty-nine percent had reflux hypersensitivity, and forty-four percent had functional heartburn. Drawn as three bars, the tallest bar is functional heartburn. Put differently, nearly three in four children with heartburn and a normal endoscopy did not have abnormal acid exposure, and most would have stayed on PPIs indefinitely without testing.

The new Rome criteria for children, published in 2026, formally add reflux hypersensitivity and reflux-negative oesophageal pain disorder, the paediatric counterpart of functional heartburn, with symptoms at least three days a week for at least two months, a normal endoscopy without eosinophilic oesophagitis, and reflux monitoring. For these children, treatment is different: limited acid suppression, and consideration of neuromodulators and behavioural therapy rather than ever-higher PPI doses.

Part fourteen. What the guidelines say.

On the empiric PPI trial, the paediatric and adult documents are aligned: four to eight weeks for typical symptoms in older children, never as a test in infants, and a response that is not conclusive. On extra-oesophageal symptoms they are also aligned: no empiric PPI as a diagnostic test. On the threshold for abnormal acid exposure they differ. The paediatric guideline uses a reflux index above ten percent under one year and above five percent over one year, without validated norms, while Lyon 2.0 uses above six percent as conclusive, below four as normal and four to six as a grey zone. On endoscopy they differ in emphasis: the paediatric guideline says not to use endoscopy alone to diagnose GERD and reserves it for complications and escalation, while the adult documents treat grade B to D oesophagitis, Barrett's and stricture as conclusive and make endoscopy the first test after a failed PPI trial. The Los Angeles classification and wireless monitoring are gaps in the paediatric document. On histology and on testing off versus on PPI, the documents are broadly aligned. The one material gap concerns functional phenotypes: in 2018, separating them fell short of consensus, while Rome V now gives paediatric criteria for them. The paediatric guideline predates both Lyon 2.0 and Rome V, and its authors were writing for a population with no normal values.

Part fifteen. Where the diagnosis goes wrong.

Several reasoning errors are common. The first is to conclude that a child who improved on a PPI has GERD. The positive likelihood ratio is only about 1.7, and placebo response and natural fluctuation explain many responses; the better test is to stop the drug and see. The second is to conclude that a normal pH study means no reflux. pH monitoring alone misses weakly acidic reflux, and in the German series impedance alone detected forty-five percent of abnormal studies; read the impedance and the symptom association too. The third is to treat grade A oesophagitis as proof. It is borderline, it is seen in healthy adults, and it needs reflux monitoring. The fourth is to try a PPI for chronic cough as a diagnostic test, which the guideline advises against. Two more errors are worth naming: performing endoscopy while the

child is still on a PPI, which heals grade A and B lesions and lowers the yield, and applying adult acid exposure thresholds to a toddler as though they were validated, when in young children the numbers are conventions agreed in symptomatic patients.

What to do differently in clinic.

For the adolescent with heartburn, a simple sequence works. First, look for alarm features such as dysphagia, bleeding, weight loss or persistent vomiting; if they are present, go straight to endoscopy with biopsies. Second, if there are none, give a PPI once daily before breakfast for four to eight weeks, for example omeprazole twenty milligrams. If she responds, stop or taper the drug, and remember that a relapse is a reason to test, not a reason to continue indefinitely. Third, if she does not respond or relapses, perform endoscopy two to four weeks off the PPI with biopsies for eosinophilic oesophagitis; grade B, C or D oesophagitis proves GERD. Fourth, if endoscopy is normal, perform twenty-four-hour pH-impedance off PPI, and read the acid exposure time, the number of reflux episodes, the baseline impedance and the symptom association together. Fifth, name the phenotype: non-erosive GERD is treated as GERD, while reflux hypersensitivity and functional heartburn are treated with limited acid suppression, neuromodulation and behavioural therapy. Before fundoplication or long-term PPI therapy in an adolescent without erosive disease, document abnormal acid exposure, which is what Lyon calls actionable GERD. And do not answer a failed first PPI trial with a second, higher-dose trial; testing answers the question, more PPI rarely does.

Return to the girl from the beginning. She should stop her PPI for two to four weeks and, if symptoms return, have an endoscopy with biopsies and then pH-impedance off treatment if the endoscopy is normal. If her acid exposure time is 5.1 percent with ninety-two reflux episodes, a baseline impedance of 1,300 ohms and a positive symptom association probability, the grey-zone result is supported by every tie-breaker and she can be treated as having GERD. If instead her acid exposure time is 2.4 percent with a symptom index of seventy percent and a normal endoscopy, she has reflux hypersensitivity, and the conversation turns from lifelong PPI to managing a sensitive oesophagus. A twelve-year-old with chronic cough and no heartburn should not be given an empiric PPI as a test. And a thirteen-year-old boy with dysphagia and food impaction needs endoscopy first, with eosinophilic oesophagitis assumed until proven otherwise.

So tomorrow in clinic, remember these numbers. An empiric PPI trial of four to eight weeks is acceptable for typical symptoms in older children, never as a test in infants. A response has a sensitivity of seventy-eight percent, a specificity of fifty-four percent and a likelihood ratio of about 1.7. Off PPI, an acid exposure time above six percent is conclusive and below four percent is physiological. More than eighty reflux episodes and a baseline impedance below 1,500 ohms support GERD. Los Angeles grade B is now enough. Erosive oesophagitis is found in about one in eight paediatric endoscopies, and confirmed Barrett's in one child in 840.

And among children with heartburn and a normal endoscopy, only about a quarter have non-erosive GERD. Above all, remember that in a child, feeling better on a PPI is the start of the diagnosis, not the end of it.